

2. (a) $F = \bar{A} + \bar{A}B + AC = \bar{A} + C$

(b) $F = \overline{\bar{A}B + \bar{A}CD + BD\bar{D}} = \overline{\bar{A}B + \bar{A}CD}$

4. $Y_3 = X_3$

$Y_2 = X_2 \oplus X_3$

$Y_1 = X_1 \oplus (MX_2 + \bar{M}Y_2)$

$Y_0 = X_0 \oplus (MX_1 + \bar{M}Y_1)$

$M=0$

$Y_3 = X_3$

$Y_2 = X_2 \oplus X_3$

$Y_1 = X_1 \oplus Y_2$

$Y_0 = X_0 \oplus Y_1$

X_3	X_2	X_1	X_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	1
0	1	0	1	0	1	1	0
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	1
1	0	0	0	1	1	1	1
1	0	0	1	1	1	1	0
1	0	1	0	1	1	0	0
1	0	1	1	1	1	0	1
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	1
1	1	1	0	1	0	1	1
1	1	1	1	1	0	1	0

$M=1$

$Y_3 = X_3$

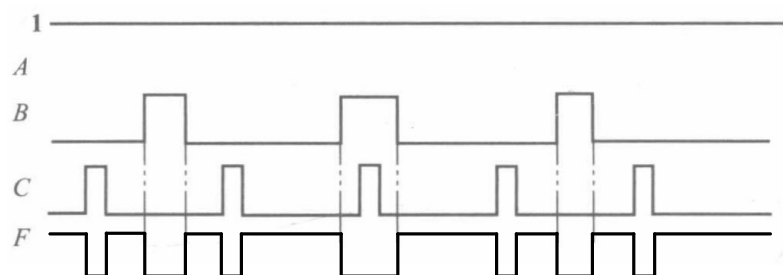
$Y_2 = X_2 \oplus X_3$

$Y_1 = X_1 \oplus X_2$

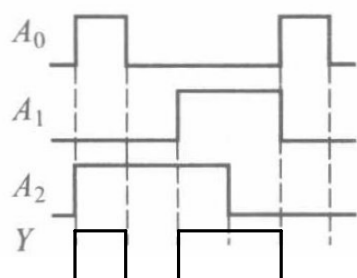
$Y_0 = X_0 \oplus X_1$

X_3	X_2	X_1	X_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

5. (a) $F = \overline{A(B+C)}$

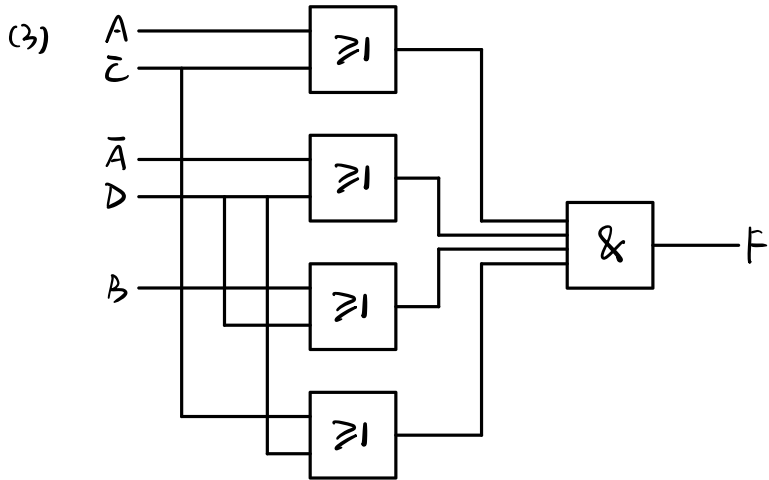
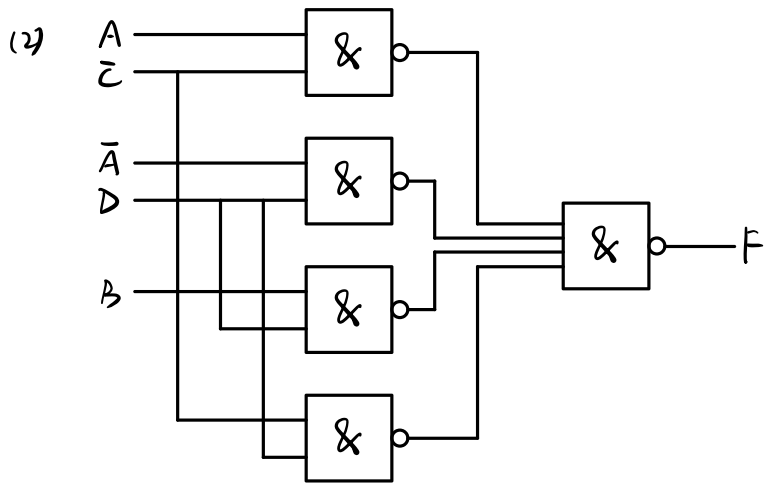
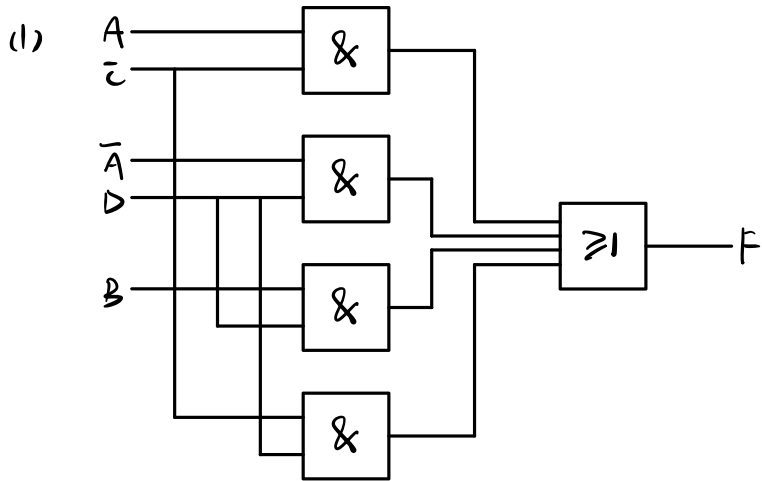


(b) $Y = \bar{A}_0 A_1 A_2 + A_0 \bar{A}_1 A_2 + \bar{A}_0 A_1 \bar{A}_2$

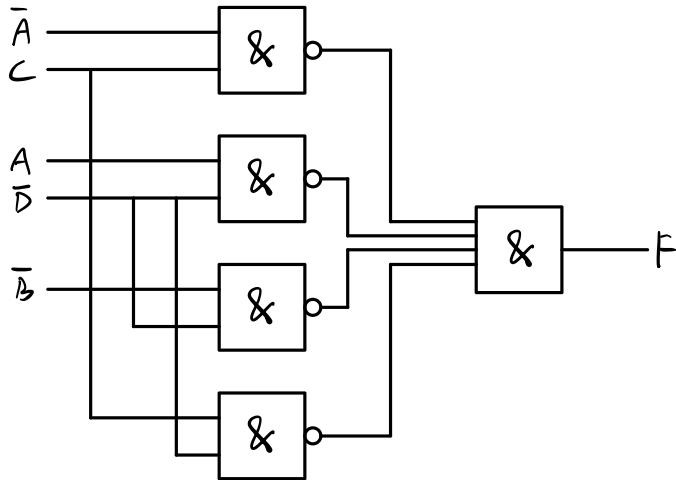
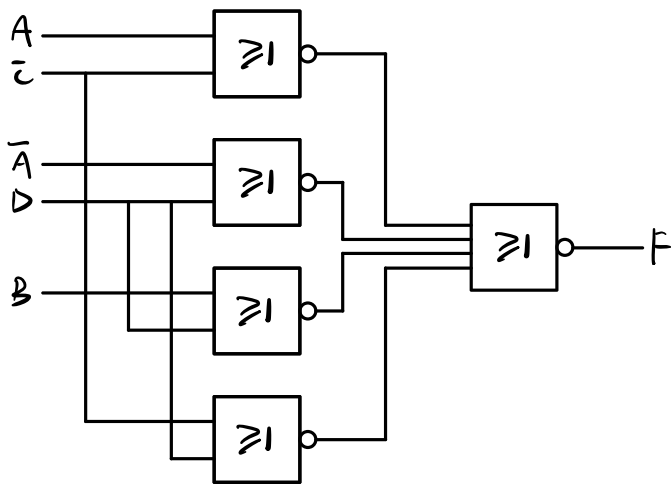


6. $F = A\bar{C} + \bar{A}D + BD + \bar{C}D = \overline{\overline{A\bar{C}} \cdot \overline{\bar{A}D} \cdot \overline{BD} \cdot \overline{\bar{C}D}} = (A + \bar{C})(\bar{A} + D)(B + D)(\bar{C} + D)$

$$= \overline{\overline{A + \bar{C}} + \overline{\bar{A} + D} + \overline{B + D} + \overline{\bar{C} + D}} = \overline{\bar{A}\bar{C} \cdot \bar{A}\bar{D} \cdot \bar{B}\bar{D} \cdot \bar{C}\bar{D}}$$



(4)



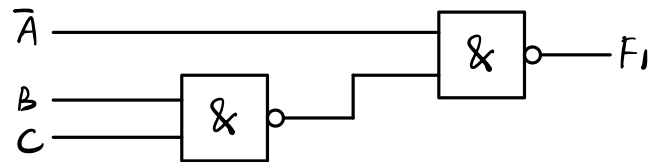
8.

A	B	C	D	F ₁	F ₂
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	0	1
0	0	1	1	0	1
0	1	0	0	0	1
0	1	0	1	0	1
0	1	1	0	1	0
0	1	1	1	1	0
1	0	0	0	1	0
1	0	0	1	1	0
1	0	1	0	X	X
1	0	1	1	X	X
1	1	0	0	X	X
1	1	0	1	X	X
1	1	1	0	X	X
1	1	1	1	X	X

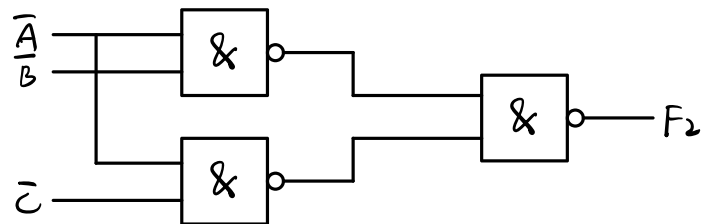
AB \ CD	00	01	11	10
00			X	1
01			X	1
11		1	X	X
10		1	X	X

AB \ CD	00	01	11	10
00	1	1	X	
01	1	1	X	
11	1		X	X
10	1		X	X

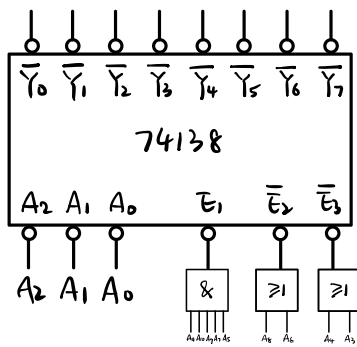
$$F_1 = A + BC = \overline{\overline{A} \cdot \overline{BC}}$$



$$F_2 = \overline{AB} + \overline{AC} = \overline{\overline{\overline{AB}} \cdot \overline{\overline{AC}}}$$



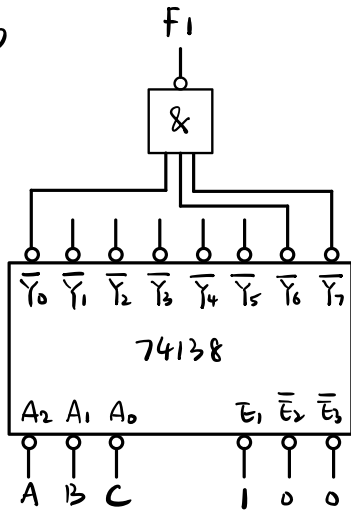
11.



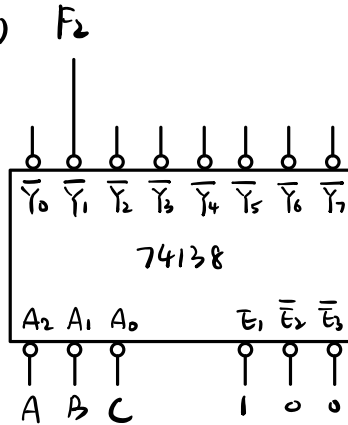
$$EA0H \sim EA7H$$

$$111010100000 \sim 111010100111$$

15. (1)



(2)



$$19. (1) W_1 = m_0 \cdot 0 + m_1 \cdot \bar{D} + m_2 \cdot D + m_3 \cdot 1$$

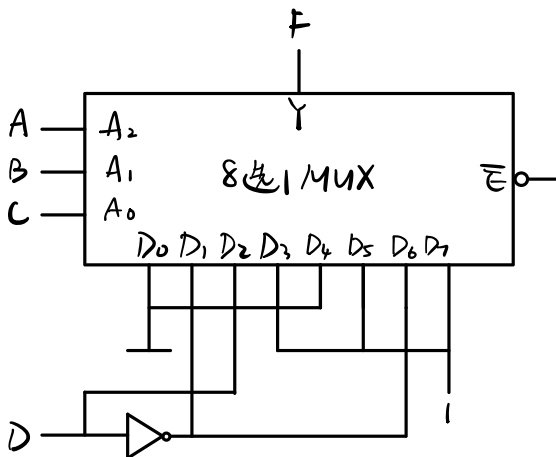
$$= \bar{B}\bar{C}\bar{D} + B\bar{C}\bar{D} + B\bar{C}D$$

$$W_2 = m_0 \cdot \bar{D} + m_1 \cdot 1 + m_2 \cdot \bar{D} + m_3 \cdot 1$$

$$= \bar{B}\bar{C} + B\bar{C}\bar{D} + B\bar{C}D$$

$$F = W_1 \bar{A} + W_2 A = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + A\bar{B}\bar{C} + A\bar{B}\bar{C}\bar{D} + B\bar{C} = \sum m(2, 5, 10, 11, 12, 14, 15)$$

$$(2) F = m_0 \cdot 0 + m_1 \cdot \bar{D} + m_2 \cdot D + m_3 \cdot 1 + m_4 \cdot 0 + m_5 \cdot 1 + m_6 \cdot \bar{D} + m_7 \cdot 1$$



22. (1) $F = AB + \bar{A}C + \bar{B}C = AB + C$ 不存在冒险现象.

(2) $F = (A + \bar{B} + C)(\bar{A} + \bar{B} + C)(A + B + C) = (A + C)(\bar{B} + C)$ 不存在冒险现象.